Model Interpretation Report

Best model carried forward from Question 4: LogisticRegression.
Test accuracy: 1.000; test macro F1: 1.000.
How the model makes predictions:
The classifier learns patterns that separate users into Low, Medium, and High activity groups using behavioral and demographic features.
For the selected logistic regression model, larger coefficient magnitudes indicate stronger influence on the class decision boundary.
In practice, the model mainly responds to transaction intensity, average transaction size, median transaction size, send/receive balance, and selected demographic indicators.

Most influential signals from the saved feature-importance summary:
- num__TX_Per_Month: 0.316
- num__Avg_Amount_XAF: 0.178
- num__Median_Amount_XAF: 0.150
- num__Send_Receive_Ratio: 0.135
- num__Household size: 0.133
- cat__Education level_Bachelor: 0.122
- num__Total_OUT: 0.116
- num__Total_Transactions: 0.115
- num__Financial_Transactions: 0.110
- num__Total_IN_Volume_XAF: 0.106

Interpretation:
Users with higher TX_Per_Month, larger transaction values, and stronger total activity tend to move toward the High activity segment.
Users with very low transaction frequency and smaller value patterns tend to be classified as Low activity.
Medium users usually sit between those extremes and often reflect moderate monthly frequency rather than isolated high-value transactions.

Example predictions are saved in example_predictions.csv and include one high-confidence case per predicted segment.
Caution: 30 demographic profiles in the 50-user build are generated placeholders, so demographic interpretations should be described as provisional.